

Assignment 3

Game Programming

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Class: BSSE-5
Subject: Game Programming

Multiplayer Game Code Architecture Design

Selected Game: Among Us

1. Game Overview

Among Us is a multiplayer online social deduction game where 4–15 players participate in a match. Most players act as Crewmates and complete tasks, while one or more players act as Impostors who try to eliminate Crewmates without being detected. The game requires real-time networking, player synchronization, task management, voting mechanics, and game state management.

2. High-Level Architecture

Player System

Handles player profiles, movement, tasks, status, and roles.

Networking System

Synchronizes player actions between clients and server.

Matchmaking System

Creates lobbies, manages room joining, and starts matches.

Task System

Assigns and tracks completion of tasks.

Voting & Meeting System

Manages emergency meetings and voting.

Game State Management

Controls game phases such as Lobby, Playing, Meeting, and End Game.

UI System

Displays menus, tasks, voting screens, and notifications.

Audio System

Handles sound effects and game feedback.

3. Code Structure

Main Classes and Responsibilities

GameManager

Controls overall game flow and game states.

PlayerManager

Stores all active players and player information.

Player

Represents an individual player and their behavior.

NetworkManager

Handles communication between server and clients.

LobbyManager

Creates and manages game lobbies.

TaskManager

Assigns and validates player tasks.

VotingManager

Handles meetings and vote calculations.

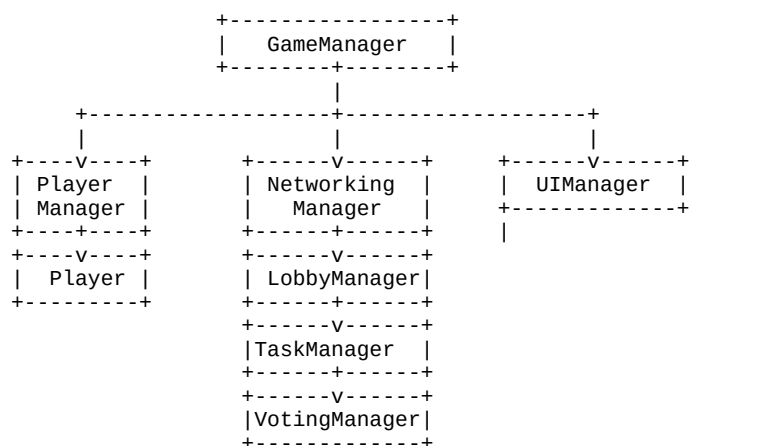
UIManager

Updates user interface elements.

AudioManager

Controls game audio.

System Communication Diagram



Communication Flow:

Players send actions to the NetworkManager. The NetworkManager validates and forwards data to GameManager. GameManager updates the game state and notifies TaskManager, VotingManager, and UIManager. The updated information is then synchronized across all connected players.

4. Design Decisions

This architecture follows a modular design approach. Each system has a single responsibility, which improves maintainability, scalability, and testing. The GameManager acts as the central controller, reducing dependency between systems. Networking is isolated in a dedicated module, making it easier to support multiplayer synchronization. The architecture is suitable for games like Among Us because it supports real-time communication, lobby management, voting mechanics, and task tracking while keeping the code organized.

Advantages of this architecture include:

- Easy maintenance and debugging
- Better scalability for future features
- Clear separation of responsibilities
- Improved multiplayer synchronization
- Reusable and modular code structure